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<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-
fit=cover">
<title>Olympiad City</title>
<style>
:root{--bg:#f4f6fb;--card:#fff;--tx:#1b2333;--mut:#6b7488;--ac:#3b6cf6;--
ok:#1f9d55;--bad:#d64545;--bd:#dfe4ee;box-sizing:border-box;padding-top:env(safe-
area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px)}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#10141f;--
card:#1a2030;--tx:#eef1f8;--mut:#9aa4bb;--ac:#6d93ff;--ok:#3ecf7e;--bad:#ff6b6b;--
bd:#2b3348;color-scheme:dark}}
:root[data-theme="dark"]{--bg:#10141f;--card:#1a2030;--tx:#eef1f8;--mut:#9aa4bb;--
ac:#6d93ff;--ok:#3ecf7e;--bad:#ff6b6b;--bd:#2b3348;color-scheme:dark}
*{box-sizing:border-box}body{margin:0;background:var(--bg);color:var(--
tx);font:16px/1.45 system-ui,-apple-system,Segoe UI,Roboto,sans-serif}
#app{max-width:520px;margin:0 auto;padding:14px}
h1{font-size:22px;margin:2px 0}h2{font-size:19px;margin:8px 0}
.card{background:var(--card);border:1px solid var(--bd);border-
radius:16px;padding:14px;margin:10px 0}
.row{display:flex;gap:10px;align-items:center;justify-content:space-between}
.mut{color:var(--mut);font-size:14px}.big{font-size:34px}
.btn{background:var(--ac);color:#fff;border:0;border-radius:12px;padding:12px
16px;font:inherit;font-weight:600;width:100%;margin-top:8px;cursor:pointer}
.ghost{background:transparent;color:var(--ac);border:1px solid var(--ac);border-
radius:12px;padding:10px 14px;font:inherit;cursor:pointer;margin-
top:8px;width:100%}
.o{display:block;width:100%;text-align:left;background:var(--card);color:var(--
tx);border:1px solid var(--bd);border-radius:12px;padding:12px;margin:8px
0;font:inherit;cursor:pointer}
.o.ok{border-color:var(--ok);background:color-mix(in srgb,var(--ok) 15%,var(--
card))}
.o.no{border-color:var(--bad);background:color-mix(in srgb,var(--bad) 15%,var(--
card))}
.bar{height:8px;background:var(--bd);border-radius:6px;overflow:hidden}.bar
i{display:block;height:100%;background:var(--ac);transition:width .5s}
.city{position:relative;border-radius:18px;overflow:hidden;padding:10px;margin-
top:10px;background:linear-gradient(#7cc8ff 0,#d6f0ff 22%,#9bd67a 22%,#62b653
100%);border:1px solid var(--bd)}
.sun{position:absolute;right:14px;top:6px;font-size:30px;animation:spin 20s linear
infinite}
.cl{position:absolute;top:10px;left:-40px;font-size:28px;animation:drift 25s
linear infinite}
@keyframes drift{to{left:110%}}@keyframes spin{to{transform:rotate(360deg)}}
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.say{position:relative;background:#fff;color:#1b2333;border-radius:14px;padding:8px 10px;font-size:14px;margin:36px 0 10px;box-shadow:0 2px 6px #0003}
.zones{display:grid;grid-template-columns:1fr 1fr;gap:10px}
.z{position:relative;border-radius:14px;padding:8px;color:#1b2333;border:3px solid #ffffffb3;cursor:pointer;min-height:150px}
.z h3{font-size:13px;margin:0 0 6px}
.plots{display:grid;grid-template-columns:repeat(3,1fr);gap:4px}
.pl{aspect-ratio:1;background:#ffffff73;border-radius:8px;display:grid;place-items:center;font-size:25px}
.b{display:inline-block;animation:pop .7s}
@keyframes pop{0%{transform:scale(0) translateY(14px)}70%{transform:scale(1.35)}100%{transform:scale(1)}}
.fog{position:absolute;inset:0;background:#eef2fae6;border-radius:11px;display:grid;place-items:center;text-align:center;font-size:12px;color:#445;padding:8px}
.ck{display:block;padding:8px 0;border-top:1px solid var(--bd)}
.warn{animation:sh .5s 3}@keyframes sh{25%{transform:translateX(-6px)}75%{transform:translateX(6px)}}
</style></head><body><div id="app"></div>
<script>
const D=[
{n:"Market Square",e:"🏪",learn:"Elasticity = %ΔQ ÷ %ΔP. If |E| > 1 (elastic), a price cut raises revenue. MR = ΔTR ÷ ΔQ. A firm maximizes profit where MR = MC. Economic profit = revenue − explicit costs − opportunity costs.",q:[
["A price cut increases total revenue. Demand is best described as:",["Elastic","Inelastic","Unit elastic"],0,"Revenue rises when price falls only if quantity rises by a bigger percentage, so demand is elastic."],
["Quantity demanded rises 5% when price falls 10%. Demand is:",["Elastic","Inelastic","Perfectly elastic"],1,"E = 5 ÷ 10 = 0.5, which is below 1."],
["With perfectly elastic supply, a new tax is paid entirely by:",["The seller","The buyer","Nobody"],1,"Sellers won’t accept less, so the price rises by the full tax."],
["MR = 150 − 10Q and MC = 3Q² − 20Q + 73. The profit-maximizing output is closest to:",["7","8","11"],0,"Set MR = MC: 3Q² − 10Q − 77 = 0, so Q = 7."],
["Grey quit a $35,000 job and put $70,000 into his business (it could earn $43,000 elsewhere). Revenue 200,000, raw materials 15,000, wages 20,000, interest 3,000. Economic profit?",["$14,000","$84,000","$132,000"],1,"Accounting profit = 162,000. Subtract 35,000 + 43,000 of opportunity cost: 84,000."],
["Budget $110. Chicken $7.50/kg, lamb $10/kg. Chicken quantity is 35% below lamb. Chicken (kg)?",["4.8","5.1","2.6"],0,"c = 0.65l, so 7.5(0.65l) + 10l = 110, l ≈ 7.39 and c ≈ 4.8."]]},
{n:"Macro Tower",e:"🏙️",learn:"CPI = cost of the base basket at current prices ÷ its cost at base prices × 100. GDP = C + I + G + (X − M). Multiplier = 1 ÷ (1 − MPC × (1 − t)). Unemployment rate = unemployed ÷ labor force.",q:[
["Base: 25 apples at $1.00 and 5 shirts at $20. Current prices: apples $1.25,
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shirts $20.50 (same quantities). The CPI is closest to:",
["103.57","107.00","113.75"],1,"Base basket = 125. Current = 31.25 + 102.5 = 133.75. CPI = 107."],
["MPC is 86% and the tax rate is 30%. Government spending rises $2 billion. Income rises by closest to:",["$2.0","$2.5","$5.0"],2,"Multiplier = 1 ÷ (1 − 0.86 × 0.7) ≈ 2.51, and 2.51 × 2 ≈ 5.0."],
["Consumption 30.0, statistical discrepancy 1.0, capital consumption 3.0, government 7.6, imports 3.4, investment 8.0, exports 3.0. National income and GDP are:",["41.2 and 45.2","45.2 and 41.2","46.2 and 42.2"],0,"GDP = 30 + 8 + 7.6 + (3 − 3.4) = 45.2. National income = 45.2 − 3 − 1 = 41.2."],
["The most likely initial effect of demand-pull inflation is a rise in:",["The price level, with real GDP falling","The price level, with real GDP rising","Government spending, then less money"],1,"Higher demand pulls up prices and, at first, output."],
["Government borrowing to finance extra spending most likely increases:",["Private investment","The real interest rate","The supply of loanable funds"],1,"Extra demand for loanable funds pushes the real interest rate up (crowding out)."],
["The labor force participation ratio falls, employment is unchanged, working-age population is constant. The unemployment rate will:",["Increase","Decrease","Stay unchanged"],1,"People leave the labor force, so the number of unemployed falls while employment stays the same."]]},
{n:"Finance Quarter",e:"🏦",learn:"WACC = E/V·Re + D/V·Rd·(1 − t). g = ROE × (1 − payout). Justified forward P/E = payout ÷ (r − g). Gordon model: P = D1 ÷ (r − g). Tax-equivalent yield = yield ÷ (1 − t). Time value: FV = PV × (1 + r)^n.",q:[
["Debt-to-equity 50%, tax 40%, cost of debt 8%, cost of equity 13%. WACC is closest to:",["7.5%","8.9%","10.3%"],2,"D/V = 1/3, E/V = 2/3: (1/3)(8)(0.6) + (2/3)(13) ≈ 10.27%."],
["Forward EPS $3.00, ROE 12.5%, payout 60%, required return 10%. Justified value per share?",["$36","$48","$72"],0,"g = 12.5% × 0.4 = 5%. P/E = 0.6 ÷ 0.05 = 12, and 12 × 3 = $36."],
["Last dividend $2.00, growing 8% for two years. Required return 11%. Stock sold for $75 after two years. Maximum price today?",
["$58.68","$64.71","$66.63"],1,"2.16 ÷ 1.11 + (2.3328 + 75) ÷ 1.11² ≈ 64.71."],
["A tax-exempt bond yields 4.5% and a taxable bond yields 6.0%. Which marginal tax rate makes you indifferent?",["25.0%","28.6%","33.3%"],0,"4.5 ÷ (1 − t) = 6, so t = 25%."],
["How long for $2,500 to grow to $7,500 at 6% compounded annually?",["11.9 years","18.9 years","21.3 years"],1,"n = ln 3 ÷ ln 1.06 ≈ 18.9."],
["Comprehensive income $150m, other comprehensive income $40m, 30m shares, price $40. P/E on a net income basis?",["10.91","8","3.667"],0,"Net income = 110m, EPS = 3.667, P/E = 40 ÷ 3.667 ≈ 10.91."]]},
{n:"Portfolio Park",e:"📈",learn:"CAPM: E(R) = Rf + β(Rm − Rf). Sharpe = (Rp − Rf) ÷ σ. Correlation = Cov ÷ (σ1σ2). Two-asset variance = w1²σ1² + w2²σ2² + 2w1w2Cov. Call payoff = max(S − K, 0). Put payoff = max(K − S, 0).",q:[
["Beta 0.65, risk-free rate 3%, market return 9%. Expected return is closest to:",
["3.9%","6.9%","10.8%"],1,"3 + 0.65 × 6 = 6.9%."],
["Rf = 4%. Portfolio D: 10% return, 20% σ. E: 18%, 15%. F: 6%, 3%. Best Sharpe

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ratio?",["D","F","E"],2,"D = 0.30, F = 0.67, E = 0.93."],
["Variances 15.5 and 22.3, covariance 8.65. The correlation is closest to:",
["0.025","0.388","0.465"],2,"8.65 ÷ √(15.5 × 22.3) ≈ 0.465."],
["Returns: 15% chance of −12%, 60% chance of 11%, 25% chance of 18%. The variance
is closest to:",["44.36","50.94","88.71"],2,"Mean = 9.3. Variance = 0.15(21.3²) +
0.6(1.7²) + 0.25(8.7²) ≈ 88.71."],
["Weights 40% and 60%, standard deviations 7% and 12%, covariance −0.004.
Portfolio standard deviation?",["6.36%","6.56%","6.10%"],0,"Variance = 0.000784 +
0.005184 − 0.00192 = 0.004048, so σ ≈ 6.36%."],
["A put has a strike of $35 and the stock ends at $33.50. The payoff is closest
to:",["$0","$1.48","$1.50"],2,"max(35 − 33.5, 0) = 1.50."]]}];

const COL=["#ffd9a8","#cfe3ff","#d5f5c8","#f2d4ff"],CR=["Price war","Inflation
spike","Bank run","Market crash"],
BLD=[["🏧","🥖","👘","🏬","🏛️","🎡"],["🏢","🏗️","🌆","🏫","🏥","🗼"],
["🏦","💳","🏧","🏰","🏠","🏰"],["📈","🎪","🎠","🌳","⛲","🚘"]],
BS=[{n:"The 2007–2009 Crisis",e:"💥",t:"Explain the causes, key events and lessons
of the Global Financial Crisis. Evaluate the policy response and the regulatory
failures.",p:["Subprime mortgages given to weak borrowers","Securitization: MBS
and CDOs spread the risk","High financial leverage in banks","Housing bubble
burst, defaults rose","Lehman Brothers collapsed in September 2008","Central bank
rate cuts and bailouts (e.g. TARP)","Regulatory failures: rating agencies, shadow
banking","Lessons: capital rules, stress tests, Dodd-Frank"]},
{n:"Securitization",e:"🧩",t:"Describe the securitization process in detail (MBS,
ABS).",p:["Originator makes loans (mortgages, car loans, cards)","Loans are pooled
together","Pool is sold to a special purpose vehicle (SPV)","SPV issues bonds in
tranches (senior to equity)","Credit enhancement and rating agencies","Investors
receive the pool’s cash flows","Risks: prepayment and default","Benefits:
liquidity and funding for lenders"]},
{n:"Crowdsourcing Model",e:"👥",t:"Describe a crowdsourcing business model in
detail. Give three examples.",p:["A platform connects a company and a
crowd","Crowd gives ideas, work, content or money","Incentives: rewards,
recognition, equity","Revenue: fees or commissions","Quality control and
moderation","Example 1 (e.g. Wikipedia or Waze)","Example 2 (e.g.
Kickstarter)","Example 3 (e.g. Upwork or Threadless)"]}];
let S={done:{},wrong:{},xp:0,best:0,boss:{},streak:0,last:""};
try{const t=localStorage.getItem("oc2");if(t)S=JSON.parse(t)}catch(e){}
const save=()=>{try{localStorage.setItem("oc2",JSON.stringify(S))}catch(e){}};
const td=new Date().toDateString();
if(S.last!=td){S.streak=(S.last&&(new Date(td)-new Date(S.last))/864e5==1)?
S.streak+1:1;S.last=td;save()}
const $=h=>document.getElementById("app").innerHTML=h;
const cnt=k=>D[k].q.filter((_,j)=>S.done[k+"-"+j]).length,open=k=>k==0||cnt(k-
1)>=3,
total=()=>D.reduce((a,_,k)=>a+cnt(k),0),lv=
()=>Math.floor(S.xp/30)+1,fmt=s=>Math.floor(s/60)+":"+String(s%60).padStart(2,"0");
let Q=[],i=0,sc=0,wrn=0,mode="",tm=null,left=0,ans=false;
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const hint=t=>t==0?"Welcome, Mayor! Our land is empty. Tap Market Square and let's build.":t<6?"Nice start! Every correct answer raises a new building.":t<12?"The city is growing. Watch out for crises, Mayor!":"A model economy! Time for the Boss Battles below.";
function zone(k){const o=open(k);return `<div class="z" style="background:${COL[k]}" onclick="${o?`dist(${k})`:`home('⬜ Clear 3 missions in the previous district to lift the fog.')`}"><h3>${D[k].e} ${D[k].n}</h3><div class="plots">${BLD[k].map((e,j)=>`<div class="pl">${S.done[k+"-"+j]?`<span class="b">${e}</span>`:`<span style="opacity:.35">🚧</span>`}</div>`).join("")}</div>${o?"":`<div class="fog">⬜<br>Fog: solve 3 missions in the previous district</div>`}</div>`}
function home(msg){clearInterval(tm);const t=total(),wr=Object.keys(S.wrong).length;
 $(`<div class="row"><h1>🏙️ Olympiad City</h1><b>🔥 ${S.streak}</b></div>
 <div class="row mut"><span>Mayor level ${lv()}</span><span>${S.xp%30}/30 XP</span></div><div class="bar"><i style="width:${S.xp%30/30*100}%"></i></div>
 <div class="city"><span class="sun">☀️</span><span class="cl">☁️</span><div class="say">🦉 <b>Prof. Aqsha:</b> ${msg||hint(t)}</div><div class="zones">${D.map((_,k)=>zone(k)).join("")}</div></div>
 <h2>Boss battles</h2>${BS.map((b,x)=>t>=6?`<button class="ghost" onclick="boss(${x})">${b.e} ${b.n} ${S.boss[x]?"✅":""}</button>`:`<button class="ghost" disabled style="opacity:.5">🔒 ${b.n} (build 6 buildings)</button>`).join("")}
 <button class="btn" onclick="mock()">⏱️ Mock Day: 12 questions, 10 min</button>
 <button class="ghost" onclick="notebook()">📓 Mistake Notebook (${wr})</button>`)}
function dist(k){const d=D[k];$(`<button class="ghost" onclick="home()" style="width:auto">← City</button><h2>${d.e} ${d.n}</h2><div class="card"><b>🦉 Learn card</b><p>${d.learn}</p></div><p class="mut">${cnt(k)}/${d.q.length} buildings built</p><button class="btn" onclick="start(${k})">Start missions</button>`)}
function start(k){mode="d"+k;Q=D[k].q.map((_,j)=>[k,j]);go()}
function mock(){mode="mock";const a=[];D.forEach((d,k)=>d.q.forEach((_,j)=>a.push([k,j])));Q=a.sort(()=>Math.random()-.5).slice(0,12);go()}
function notebook(){const a=Object.keys(S.wrong).map(s=>s.split("-").map(Number));if(!a.length)return home("No mistakes in the notebook. Great work, Mayor!");mode="nb";Q=a;go()}
function go(){i=0;sc=0;wrn=0;show()}
function show(){ans=false;const [k,j]=Q[i],q=D[k].q[j];
 $(`<div class="row"><button class="ghost" onclick="home()" style="width:auto;margin:0">✕</button><span class="mut">${i+1}/${Q.length}</span><b>${sc} pts</b>${mode=="mock"?`<span id="t">${fmt(left)}</span>`:""}</div>
 <div class="bar" style="margin-top:8px"><i style="width:${i/Q.length*100}%"></i></div>
 <div class="card ${mode[0]=="c"?"warn":""}"><div class="mut">${mode[0]=="c"?"⚠️
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CRISIS: "+CR[k]+" · ":""}${D[k].e} ${D[k].n}</div><p><b>${q[0]}</b>
</p>${q[1].map((o,x)=>`<button class="o" onclick="pick(${x})">${"ABC"[x]}. ${o}
</button>`).join("")}
 <button class="ghost" id="sk" onclick="pick(-1)">Skip (0 pts)</button><div
id="ex"></div></div>`)}
function pick(x){if(ans)return;ans=true;const [k,j]=Q[i],q=D[k].q[j],id=k+"-
"+j,b=document.querySelectorAll(".o");
 b[q[2]].classList.add("ok");
 if(x==q[2]){sc+=3;S.done[id]=1;delete S.wrong[id]}else if(x>=0){sc-
=1;wrn++;b[x].classList.add("no");S.wrong[id]=1}
 save();document.getElementById("sk").style.display="none";
 document.getElementById("ex").innerHTML=`<p class="mut">${x==q[2]?"✅ +3 · a
building rises!":x<0?"Skipped, 0 points":"❌ -1"} ${q[3]}</p><button class="btn"
onclick="nx()">${i+1<Q.length?"Next":"Finish"}</button>`}
function nx(){i++;i<Q.length?show():end()}
function result(t,m,c){$(`<h2>${t}</h2><div class="card"><div class="big">${c||""}
</div><p>🦉 ${m}</p></div><button class="btn" onclick="home()">Back to the
city</button>`)}
function end(){clearInterval(tm);const k=+mode[1],g=Math.max(sc,0);S.xp+=g;
 if(mode=="mock"&&sc>S.best)S.best=sc;
 if(mode[0]=="c"){if(sc>=6){S.xp+=10;save();return result("City saved! 🎉","You
stopped the "+CR[k]+". +10 bonus XP!","🛡️")}
  for(let j=5;j>=0;j--)if(S.done[k+"-"+j]){delete S.done[k+"-"+j];break}
  save();return result("The crisis hit the city 💥","The "+CR[k]+" destroyed a
building. Study the Learn card and rebuild!","🔥")}
 save();
 if(mode[0]=="d"&&wrn>0){$(`<div class="card warn"><div class="big">⚠️</div>
<h2>CRISIS: ${CR[k]}!</h2><p>🦉 Mayor! Your mistakes shook the market. Answer 2
questions correctly to defend ${D[k].n}, or we lose a building!</p></div><button
class="btn" onclick="crisis(${k})">Defend the city</button>`);return}
 result("Round complete","+"+g+" XP earned. "+(sc>=Q.length*2?"Strong run!":"Check
the Mistake Notebook."),sc+" / "+Q.length*3)}
function crisis(k){mode="c"+k;Q=D[k].q.map((_,j)=>
[k,j]).sort(()=>Math.random()-.5).slice(0,2);go()}
function boss(b){const B=BS[b];$(`<button class="ghost" onclick="home()"
style="width:auto">← City</button><h2>${B.e} Boss: ${B.n}</h2><div class="card">
<p>${B.t}</p><p class="mut">Write your answer on paper first. Then tick every key
point you covered.</p>${B.p.map(x=>`<label class="ck"><input type="checkbox"> ${x}
</label>`).join("")}</div><button class="btn" onclick="fin(${b})">Finish the
battle</button>`)}
function fin(b){const c=document.querySelectorAll(".ck
input:checked").length,n=BS[b].p.length,w=c/n>=.7;if(w&&!S.boss[b])
{S.boss[b]=1;S.xp+=20}save();result(w?"Boss defeated! 👑":"The boss
survived…",w?"You covered "+c+"/"+n+" key points. +20 XP!":"You covered
"+c+"/"+n+". Aim for 70% and try again.",c+"/"+n)}
home();
</script></body></html>
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